Shakoor Shaik

647-540-1706 | shakoorshaik7@gmail.com | LinkedIn | GitHub

Education

University of Toronto — Toronto, ON
Bachelor of Science in Computer Science, Software Engineering — Dean's List 25' — *2028*
Courses: Software Design (Object-Oriented Programming, Java), Systems Programming (Unix/Linux), Data Structure Analysis, Computer Organization (Assembly), Intro to Machine Learning

Experience

Software Engineer Intern — May. 2026 – Present
Pada Guidance
- Drove 50+ beta signups at **2.5%** conversion from **1,300** early impressions by shipping core React Native/Expo features, including driving-preferences, cruise-startup latency fixes, and high-impact sprint tickets ahead of launch for startup.
- Engineered AWS + Supabase/Postgres data pipelines serving a **100K+** lane-level road-hazard dataset, enabling real-time routing, hazard reporting, and session analytics at **sub-200ms** query latency with sub-second in-drive lane hazard display.
- Cut release-cycle time by **30%** and reduced regression defects by building up CI/CD pipelines, automated test coverage, and weekly user-feedback loops, raising automated test line coverage to **85%** across core flows in the sprint process.

Data Processing Engineer Intern — Sep. 2024 – Sep. 2025
University of Toronto Aerospace Team — *Toronto, ON*
- Designed and trained **2D/3D** CNN models in Python for hyperspectral image classification on the FINCH satellite mission, and audited and corrected **40%** of errors across relational and NoSQL mission databases to ensure clean training data.
- Contributed **1,000+** lines of Python to the open-source FINCH smile-keystone distortion-correction algorithm, a preprocessing step required before any downstream science, working alongside 5+ MSc researchers through structured Git workflows, code reviews, and cross-team technical discussions.

Full-Stack Engineer Intern — Sep. 2024 – May 2025
CREATE UofT
- Shipped a production full-stack course-planning platform serving **15,000+** UofT students, architecting React front-end, Flask/Python RESTful APIs, and a normalized MySQL backend deployed in a CI/CD pipeline for continuous release.
- Supported conflict-free scheduling across full course loads by designing a normalized relational schema with smart conflict detection over indexed joins, serving real-time validation results at sub-second query latency.
- Accelerated feature release frequency **2x** by cutting deployment time **40%** through automated CI/CD pipelines, integration testing, and agile sprint practices, shipping to active users faster.

Projects

Okra | *Next.js, Typescript, IBM watsonx.ai, Supabase, PostgreSQL, Mapbox GL, Tailwind CSS, Framer Motion, Zustand*
- Built and deployed an AI-native home-care dispatch platform for elderly Toronto residents, orchestrating **3** IBM Granite agents, for Routing, Planning, and Scheduling, to automate end-to-end care coordination from intake to dispatch.
- Reduced provider-to-patient travel time **25%** by implementing geolocation-based matching and nearest-neighbor TSP route optimization, cutting total route distance across the care network versus naive sequential assignment.
- Delivered real-time multi-user coordination through Supabase Realtime sync with row-level security, surfacing live dispatch state on a 3D Mapbox GL dashboard with urgency scoring and an AI agent chatbot for natural-language queries.

SMART Air | *Java, Kotlin, Android SDK, Firebase Firestore & Auth, Jetpack Compose, JUnit, Mockito, Espresso*
- Built a full-stack Android asthma-monitoring app for **3** user personas (children, parents, providers) featuring a clinically-modeled PEF zone system (Green/Yellow/Red), real-time zone tracking, emergency flows, and gamified adherence.
- Achieved **100%** JUnit + Mockito + Espresso unit-test coverage of core business logic by applying a hybrid MVP + Manager/Singleton pattern (ZoneManager, PEFManager, MedicineManager) to decouple Firestore operations from the UI.
- Secured multi-role access through Firebase Auth with code-based account linking and Firestore row-level rules while accelerating feature delivery by **30%** via Git workflows and Agile Scrum, all written in a Java Kotlin codebase.

Wine Quality Regression | *Python, scikit-learn, NumPy, pandas, PySR*
- Engineered a stratified data pipeline with StandardScaler normalization and 5-fold cross-validation to handle class imbalance, with Local Bayesian Regression achieving best performance (MSE: **0.351**, R^2: **0.523**, Acc±1: **97.0%**)
- Authored an academic research paper conducting a systematic comparison of 5 supervised ML regression models (KNN, LBR, BFR, MLP, SR) on the UCI Wine Quality Dataset (**6,497 samples**), documenting methodology and results.
- Optimized hyperparameters across **1,650+** MLP configurations and tuned KNN/LBR neighborhood sizes, RBF kernel widths, and evolutionary SR expression trees using grid search and CV-MSE minimization

Technical Skills

Languages: Python, Java, C, Javascript/Typescript, SQL, HTML/CSS, R
Frameworks: React.js, Node.js, Express.js, p5.js, Flask, FastAPI, REST API, scikit-learn, pandas, NumPy, TensorFlow
Developer Tools: Git/Github, AWS, Firebase DB, Supabase, Google Cloud Platform, PostgreSQL, Linux/Unix, Jira